

Problem Set 3

Demand and Demand Elasticity

EC306: Intermediate Microeconomics

Instructions. Work through these in class. Show all working and interpret your numerical answers in words. Worked solutions follow at the end.

Question 1 — Slutsky decomposition, with numbers

A consumer has $u(x, y) = x^{0.5}y^{0.5}$, income $m = 80$, and initially $p_x = 2$, $p_y = 2$. The price of x falls to $p'_x = 1$.

- (a) Find the initial bundle (x_0, y_0) and the new bundle (x_1, y_1) , and hence the *total* change in x .
- (b) Using the Hicksian (utility-held-constant) approach, find the income needed at the new prices to reach the *original* utility, and the corresponding compensated bundle.
- (c) Decompose Δx into substitution and income effects. Report both numbers.
- (d) Is x normal or inferior here, and how do you know from your two effects? In two sentences, state what would have to be true of the income effect for this good to be Giffen, and why Cobb–Douglas can never produce a Giffen good.

Question 2 — Elasticity, functional form, and revenue

- (a) For the constant-elasticity demand $Q = Ap^{-\eta}$ ($A, \eta > 0$), show the own-price elasticity is $-\eta$ everywhere, and find the elasticity of *total revenue* with respect to price. State the condition on η under which a price rise raises revenue.
- (b) For the linear demand $Q = 150 - 3p$, find: (i) the unit-elastic price, (ii) the revenue-maximising price, (iii) the price elasticity at $p = 10$ and at $p = 40$, and (iv) classify each as elastic/inelastic.
- (c) A manager says “our demand curve is a straight line, so our elasticity is constant.” Using your Q2(b) results, explain in 2–3 sentences exactly why this is wrong and what *is* constant.

Question 3 — A real elasticity, found and interpreted

- (a) Locate a published estimate of the price elasticity of demand for a specific good (e.g. petrol, cigarettes, electricity, a named foodstuff) from a credible source — a government report, regulator, or peer-reviewed paper. Report the number, the good, the country/period, and full citation.
- (b) State whether short-run and long-run estimates differ for your good, and explain *why* the time horizon changes elasticity (give the mechanism, not just the direction).
- (c) Suppose a government imposes a 15% price-raising tax on your good. Using your cited elasticity, estimate the percentage change in quantity demanded and comment on who bears most of the burden.
- (d) Name one reason your back-of-envelope estimate in (c) could be badly wrong (e.g. the elasticity is local, non-linearity, the tax is large relative to the data range).

Question 4 — Cross-price and market definition

Two products A and B have estimated cross-price elasticity $\varepsilon_{AB} = +1.8$.

- (a) Interpret the sign and magnitude. Are A and B substitutes or complements, and strongly so?
- (b) A competition authority must decide whether A and B are “in the same relevant market.” Explain how ε_{AB} informs that decision and what threshold reasoning a regulator might apply.
- (c) Give a concrete real example of two products you would expect to have a large positive cross-price elasticity, and one pair with a large *negative* one. Justify each in one sentence.